Alan Hsu

Senior at Purdue University passionate about spaceflight, looking for ownership of scalable low-cost avionics.

Email: alanjhsu08@gmail.com
Phone: (858)-442-2909
Web: alanjhsu.com

EDUCATION

08/2023 – 05/2027 **Bachelor's of Science in Electrical Engineering** | Purdue University 3.97/4 GPA
- Industry Co-Op Certificate

08/2019 – 06/2023 **Thomas Jefferson High School for Science and Technology** 4.450/4 GPA

SKILLS

- Altium Designer, KiCAD, Fusion 360, NX, Solidworks, LTSpice, Ubuntu, Python, C/C++, MATLAB
- Design for survival in High Vibration, Radiation, and Thermals; Data, Link, and Power Budgeting
- Simulation of radiation environments and determination of device upset rates using OMERE 5.9

WORK EXPERIENCE

Starpath Robotics | Web: starpath.space | Hawthorne, CA

08/2025 – Present *Avionics Engineer* | *Remote, onsite summers*
- Designed and built GaN based 60 V 5 kW_{pk} FOC BLDC motor controller, battery management for Rovers 11/12, and LED drivers for IR illuminators for permanently shadowed regions (PSR)
- Performed clean sheet design from system down to PCB module of a 1 kV 275 kW (system) solar conditioning system for space applications, targeting 5 year lifespan and a 90 °C coldplate temperature
- Simulated radiation environments in OMERE to justify radiation requirements both for Low Earth Orbit and Lunar Surface products
- Performed heavy ion testing of switching controller candidate components at Texas A&M Cyclotron

05/2024 – 08/2025 *Interim Avionics Lead* | *Onsite*
- Set scope, timelines, and budgets and kicked off Starpath's Starmine Avionics program
- Collaborated with Rover Mechanical team to negotiate thermal requirements for lunar PSR capable rover
- Designed and built all avionics systems, including for Starpath's submission to NASA BTIL Level 3, Starpath's first full system integrated demo, and Starpath's first dusty TVAC test of Rover in MSFC's V20 chamber
- Collaborated with engineers from NASA GSFC to build a custom radiation tolerant iTOF sensor

06/2022 – 08/2022 *Electrical Engineering Intern* | *Onsite*

06/2023 – 08/2023
- Integrated and harnessed electronics for Rovers 3, 4, and 8, which competed in phases of NASA BTIL

PROJECT EXPERIENCE

01/2026 – Present **APEx, Purdue CubeSat Lab** | *Payload Integration Electrical Lead*
- APEx is a mission that will demonstrate a Pulsed Magnetoplasmodynamic (PMPD) thruster on a 6U CubeSat, and is competing in the University Nanosatellite Program's NS-12 cycle
- Designed a flyback converter to charge the PMPD energy storage capacitor to 400V from 12-16V input
- Created system block diagrams for Payload system power and telemetry and control PCBs
- Delegated tasks for updating documentation, designing boards, and writing software for CDR deliverable submissions and thruster demonstration to UNP review team

01/2024 – 05/2024 **FlaC-Sat, Purdue CubeSat Lab** | *FEMTA Payload Subteam*
- FLaC-Sat was a mission to test Purdue's FEMTA UPW micropropulsion thruster concept on a 6U CubeSat, and competed in the University Nanosatellite Program's NS-11 cycle
- Developed microcontroller based control system for thrust control and temperature management of UPW propellant

08/2023 – 05/2024 **Purdue Solar Racing** | *Solar Electrical Subteam*
- Prototyped a low cost high-voltage MPPT boost converter for Lux, PSR's submission to ASC 2024

09/2021 – 06/2023 **TJ UAV Club** | *Electronics Lead, Competition Team Captain*
- Led electrical development for Avalon X, a fixed wing aircraft that competed in the 2023 SUAS competition
- Designed and built two iterations of the aircraft's electronics bay and a self-stabilizing camera gimbal

09/2021 – 06/2023 **TJ Space Program** | *TJREVERB Technical Lead*
- Led electrical development for TJREVERB, a 2U CubeSat deployed on December 29, 2022
- Designed and built a Raspberry Pi and Iridium-based CubeSat bus to provide a low cost and easy to use platform for future missions
- Coached underclassmen in mission and hardware design